

Module 5: Scientific communication and assessment

Instructional time: 30 min (10 knowledge check, 20 capstone). Bloom: Evaluate, Create. Notebook: none. Ships as: document only.

1. Overview

Modules 1 to 4 produced a set of results: a verified cohort, an integrated object with four compartments, a multi-omic interpretation, and a cross-resource comparison. Module 5 asks what can honestly be claimed from work of that kind, and then asks you to apply the same approach to a biological question of your own.

Key components. Three principles of fidelity that govern how a computational result becomes a scientific claim; a capstone in which you design an analysis for a new question using Common Fund resources; and a ten-item knowledge check.

Learning objectives. After this module you will be able to:

- Apply the principles of data, method and context fidelity to a computational result.
- Design an analysis that carries the approach used in this course to a new biological question.
- Identify which Common Fund resources bear on a question and what each one contributes.

2. Three principles of fidelity

A computational result becomes a scientific claim through three commitments. Each one is visible in the work you have just done.

Fidelity of data: report at the granularity the analysis produced

An analysis produces its result at a particular resolution, and the report should not exceed it.

Module 4 tested whether pathways concordant between the disease and aging contrasts shared enough of their driving genes to be described as one mechanism. Leading-edge clustering merged none of the five. The result is therefore five separate concordant pathways, grouped only by direction: cholesterol biosynthesis and SREBF activation down, EMT, ECM organization and keratinization up.

Describing those five as "sterol synthesis and epithelial remodeling" would be more satisfying to write, and it would assert coordinated regulation that the leading-edge overlap does not support. The same applies in the aging and flight comparison, where clustering merged exactly one pair of terms and left oxidative phosphorylation, fatty-acid beta-oxidation and mitochondrial protein import distinct. All four are mitochondrial, and shared localization is not evidence of shared regulation. "Mitochondrial metabolism is down" is a fair summary sentence; it is not a mechanism claim.

Fidelity of method: ensure reproducibility of research and findings

A result that cannot be regenerated is an anecdote. Reproducibility here is specific and checkable.

Name the exact records. HuBMAP publishes a processed derivative for each primary dataset, and Donor_1 has two published derivatives; this course pins one and says which. Cite the primary DOI, state the release analyzed, and a reader can return to your starting point.

Record the decisions the result depends on. Reference levels in this course are set explicitly, and that is not a formality: alphabetical factor ordering would have inverted three of the four contrasts in Module 4. Quality-control thresholds, gene-set libraries, random seeds and the number of permutations all belong in the record for the same reason.

Ship the artifacts a claim rests on. Every number in these documents is read from a committed table rather than transcribed, so a reader can check any figure against the file that produced it.

Fidelity of context: bound interpretation to evidence and disclose limitations

A claim carries the conditions under which it holds.

Distinguish a negative result from an uninterpretable one. In the flight contrast every pathway that passed threshold moved in the same direction. That unanimity is a warning about the contrast, not a finding about spaceflight. Three explanations were tested and rejected, and the honest report is that the contrast cannot answer a directional question, which is a different statement from finding no effect.

Keep the methods result separate from the biology result. The calibration checks in Module 4 passed convincingly. A sound pipeline pointed at a contrast that cannot answer the question is still sound, and still cannot answer the question.

State the conditions plainly. The HuBMAP age comparison is two donors against two, and is presented as a design illustration rather than a powered result. Disclosing that is what allows a reader to weigh it correctly.

3. Assessment

Module 5 is assessed in 30 minutes across two instruments, taken in this order.

Knowledge check, 10 minutes. Ten single-answer items covering Modules 1 to 4, one point each. The same items serve as pre-test and post-test, so that gain is measurable. It comes first because it asks what the course established, and answering it is the fastest way to find which parts of Modules 1 to 4 you would want to revisit before designing something of your own.

Capstone, 20 minutes. An abstract describing how you would apply the approach from this course to a different biological question, using data you locate yourself. Common Fund resources are strongly encouraged, because part of the exercise is finding out what the ecosystem holds and which parts of it bear on your question. It is the longer half because locating candidate data and establishing whether its metadata supports your contrast is the work, and it is the part that most resembles research.

Both instruments follow this document. Answer keys and grading notes are held in a separate instructor document.

4. Knowledge check

The same 10 items serve as pre-test and post-test, so that gain is measurable.

Ten minutes, one letter per item, one point each. Items 1 to 4 can be answered from the screencasts and the module documents. Items 5 to 10 test what working the notebooks teaches.

1. Your download manifest records a size and a checksum for every required file, and you check them before running anything. What does that check establish?

- A. That the data are of high enough quality to answer the biological question.
- B. That each file was retrieved from the fastest available mirror for its resource.
- C. That the files on disk are the files the analysis was written against.
- D. That the processed derivatives are equivalent to the primary datasets they came from.

2. You need lung single-nucleus data and do not know which repositories hold it. What does the CFDE discovery layer provide?

- A. Harmonized expression matrices drawn from every Common Fund repository, ready to download.
- B. A search across resources showing which repositories hold matching data, retrieved separately.
- C. A curated recommendation of the single best repository for any given tissue and question type.
- D. Direct programmatic access to all registered repositories through one shared set of credentials.

3. After batch correction across donors, the mean foreign-donor nearest-neighbor fraction rises from 0.394 to 0.685 and the donor-colored UMAP shows donors overlapping. What has the correction changed?

- A. The expression values, so the donor effect is removed before any differential testing.
- B. The cell-type labels, which are recomputed in the corrected space and replace the originals.
- C. The variance of every gene, so donor is no longer a factor in any downstream model.
- D. The embedding used for neighbors and clustering; the expression matrix is unchanged.

4. A QC notebook sets `max_mito_percent = 5` for snRNA-seq lung data. What does this course require before that threshold is applied?

- A. Nothing further, because a widely used value needs no support in the analysis that uses it.
- B. A justification for this assay, or a sensitivity check on the cutoff.
- C. Removal of the filter, because nuclear preparations contain no mitochondrial transcripts.
- D. Re-tuning per donor until every donor retains a comparable number of nuclei.

5. An Azimuth label is three times more frequent in Donor_A than Donor_B after QC. What can you conclude?

- A. Composition differs; expression differences stay undetermined without a within-label contrast.
- B. Cells carrying that label are transcriptionally more active in Donor_A than they are in Donor_B.
- C. Harmony integration failed, because correct integration equalizes label frequencies per donor.
- D. Donor_B failed quality control and ought to be dropped before any downstream analysis is run.

6. Enrichment is run on a short gene list against the full library background rather than the measured gene universe. What is the consequence?

- A. Terms become harder to detect, so genuinely enriched pathways are missed from the result.
- B. Only the direction of enrichment changes, while the significance values remain unaffected.
- C. Significance is inflated, because the background includes genes the assay never measured.
- D. The result is invalid unless the input gene list contains at least five hundred entries there.

7. In the paired multiome data, every factor explains variance in one modality and close to none in the other. What has the integration shown?

- A. No shared structure was found; the factors describe each modality separately.
- B. The two modalities are perfectly integrated, since each factor has a clear assignment.
- C. The factor count was set too low, so shared structure exists but was not recovered here.
- D. The ATAC modality failed quality control and should be excluded from the analysis.

8. Why does this course compare pathway enrichment scores across resources rather than expression values?

- A. Enrichment scores are larger numbers, so differences between resources are easier to see.
- B. Expression values are proprietary in some resources and cannot be redistributed openly.
- C. Pathway scores require no normalization, whereas expression values need batch correction first.
- D. Native units differ by platform and species; a within-resource contrast shares one design.

9. Two contrasts share 1,185 pathways, and you want to know whether the ones passing threshold in both agree in direction more often than chance. What does a Spearman correlation across all 1,185 enrichment scores tell you?

- A. Little, because it averages over all pathways rather than testing the set that passed.
- B. It answers the question, since a positive coefficient means the passing pathways agree.
- C. It answers it more reliably than a subset test, because more pathways means more power.
- D. It answers it whenever the coefficient reaches conventional statistical significance.

10. Two pathways pass threshold in the same direction in both contrasts, and leading-edge clustering leaves them in separate clusters. How should they be reported?

- A. As one mechanism, since agreeing direction in two independent contrasts is shared regulation.
- B. As a single block named for the shared process, with the member pathways listed underneath it.
- C. As two separate concordant pathways, grouped only by direction; the clustering did not merge.
- D. As one pathway plus one artifact, since two terms for the same process indicates redundancy.

Item bank note for editors. Items 1 to 3 depend on the Module 1 screencast and the "Getting the data" section of the Module 1 companion. Do not remove either without re-basing those items.

5. Capstone exercise

Time. 20 minutes. **Bloom.** Create. **Format.** A short abstract. No word count.

This course followed one approach: locate data through Common Fund discovery surfaces, bring it to a local environment, define a contrast inside each resource, and compare only what can honestly travel between them. The capstone asks you to show that you can carry that approach to a question of your own.

You are designing an analysis, not running one.

The task

Write a brief abstract for a study that applies the approach from Modules 1 to 4 to a different biological problem. Any organ, condition, or model system is fine.

Your abstract should make clear:

- **The question**, stated so that a contrast can answer it.
- **The data**, naming the resources you would use and what each one contributes.
- **The design**, meaning the case and reference groups in each resource.
- **What you would compare** across resources, and what you would not.
- **What would limit the interpretation**, and what you would report so a reader can see that limit.

Length is up to you. A paragraph that covers those five points well is worth more than a page that covers them loosely.

Finding your data

Locating the data is part of the exercise, and it is the part that most resembles real work.

Start at the CFDE discovery surfaces you used in Module 1 and search on your tissue or condition. Common Fund resources are strongly encouraged here, not because external data is disqualified, but because working out which programs hold what is one of the more durable things this course can give you. You will likely find resources you did not know existed.

For each candidate resource, establish:

1. What it actually contains: bulk or single cell, which tissues, how many donors, and what is recorded about those donors.
2. Whether the metadata supports the contrast you want. A resource with the right tissue but no record of the variable you care about cannot answer your question.
3. What you could obtain without special access, and how large it would be.

If your question needs data no Common Fund program holds, say so and name the external source. That is what Module 4 does for fibrotic human lung, and stating it plainly is better practice than stretching a resource to cover a question it was not built for.

Prompts to work through first

These are not graded. They are the questions that make the abstract writable.

4. Does one resource answer your question, or does it take two? If two, what is the shared quantity that lets them speak to each other?
5. Where is your design most likely to be underpowered, and what would you report so a reader sees it?

6. If your two resources disagreed, what would distinguish a biological difference from a design or annotation artifact?
7. Which of the three fidelity principles would be hardest to satisfy in your study, and why?

Scoring rubric (100 points)

Element	Points	What earns full marks
Question	20	Answerable by a contrast, not a description. A specific tissue or system.
Data and discovery	30	Names the resources and what each contributes. Shows the candidate data was actually checked for tissue, scale and metadata rather than assumed. Names external sources honestly where the Common Fund does not cover the question.
Design	20	Case and reference groups specified for each resource.
What travels	15	Names the comparable quantity and the quantity that must not be compared, with a reason.
Limits	15	Identifies a real limitation and says what would be reported so a reader can weigh it.

Full marks do not require a novel study. They require a design whose limits are visible in the design itself.